

Date: 2026-06-04
Request ID: 100048622

Microwave					
Element(s)	Co				
SOP#					
Acid Cocktail	9 mL HCl, 3 mL HNO3, 6 mL H2SO4				
Rack	7 or 20				
Vessel	Quartz or PTFE				
Stir	90%				
Time (min)	Power (W)	T1 (°C)	T2 (°C)	P (bar)	
15	1200	180	65	110	
15	1500	240	65	110	
30	1500	240	65	110	
sample(s)	weight (g)	volume (mL)			
200128227_1	0.1071	250			
200128227_2	0.1028	250			
200128227_3	0.1002	250			
200128227_4	0.1020	250			
200128228_1	0.1048	250			
200128228_2	0.1056	250			
200128228_3	0.1063	250			
200128228_4	0.1023	250			
200128229_1	0.1056	250			
200128229_2	0.1033	250			
200128229_3	0.1024	250			
200128229_4	0.1014	250			
200128230_1	0.1053	250			
200128230_2	0.1072	250			
200128230_3	0.1063	250			
200128230_4	0.1061	250			
200128231_1	0.1037	250			
200128231_2	0.1044	250			
200128231_3	0.1044	250			
200128231_4	0.1007	250			
200128232_1	0.1066	250			
200128232_2	0.1044	250			
200128232_3	0.1012	250			
200128232_4	0.1034	250			
200128233_1	0.1030	250			
200128233_2	0.1061	250			
200128233_3	0.1006	250			
200128233_4	0.1017	250			
200128234_1	0.1031	250			
200128234_2	0.1058	250			
200128234_3	0.1064	250			
200128234_4	0.1028	250			
200128235_1	0.1057	250			
200128235_2	0.1054	250			
200128235_3	0.1010	250			
200128235_4	0.1054	250			
200128236_1	0.1074	250			
200128236_2	0.1039	250			
200128236_3	0.1065	250			
200128236_4	0.1016	250			
200128237_1	0.1069	250			
200128237_2	0.1033	250			
200128237_3	0.1006	250			
200128237_4	0.1056	250			

Katanax		
Element(s)	Ti	
SOP#		
Acid Cocktail	30 mL conc. HCl	
sample(s)	weight (g)	volume (mL)
200128227_1	0.2057	250
200128227_2	0.2022	250
200128227_3	0.3073	250
200128227_4	0.3034	250
200128228_1	0.2054	250
200128228_2	0.2045	250
200128228_3	0.3037	250
200128228_4	0.3035	250
200128229_1	0.2026	250
200128229_2	0.2047	250
200128229_3	0.3015	250
200128229_4	0.3052	250
200128230_1	0.2057	250
200128230_2	0.2045	250
200128230_3	0.3040	250
200128230_4	0.3059	250
200128231_1	0.2070	250
200128231_2	0.2083	250
200128231_3	0.3035	250
200128231_4	0.3028	250

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128227	31.2736	33.2893	33.2403	2.43	0.9757

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128227_1	100/5	1.894	90625.06	9.06
200128227_2	100/5	1.809	90178.55	9.02
200128227_3	50/1	0.715	91418.93	9.14
200128227_4	50/1	0.749	94076.13	9.41

co result: 91574.66 ppm

co result: 9.16 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128227_1	50/6	2.038	21155.17	2.12
200128227_2	50/6	1.998	21098.95	2.11
200128227_3	50/1	0.492	20511.64	2.05
200128227_4	50/1	0.479	20226.36	2.02

ti result: 20748.03 ppm

ti result: 2.07 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
Request ID: 100048622

LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128228	26.5987	28.6155	28.5296	4.26	0.9574

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128228_1	100/5	2.078	103551.72	10.36
200128228_2	100/5	2.067	102223.23	10.22
200128228_3	50/1	0.847	104031.10	10.40
200128228_4	50/1	0.831	106056.78	10.61

co result: 103965.71 ppm

co result: 10.40 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128228_1	50/5	2.001	25438.40	2.54
200128228_2	50/5	2.011	25678.04	2.57
200128228_3	50/1	0.597	25665.08	2.57
200128228_4	50/1	0.600	25811.05	2.58

ti result: 25648.14 ppm

ti result: 2.56 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128229	26.8168	28.8552	28.7743	3.97	0.9603

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128229_1	100/4	1.800	110936.96	11.09
200128229_2	100/4	1.763	111075.85	11.11
200128229_3	50/1	0.870	110590.28	11.06
200128229_4	50/1	0.850	109113.53	10.91

co result: 110429.15 ppm

co result: 11.04 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128229_1	50/4	1.978	31770.53	3.18
200128229_2	50/4	1.999	31778.44	3.18
200128229_3	50/1	0.740	31947.88	3.19
200128229_4	50/1	0.750	31987.06	3.20

ti result: 31870.98 ppm

ti result: 3.19 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128230	32.6958	34.7245	34.6267	4.82	0.9518

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128230_1	100/4	1.969	122787.85	12.28
200128230_2	100/4	2.001	122571.74	12.26
200128230_4	50/1	1.054	130464.78	13.05

co result: 125274.79 ppm

co result: 12.53 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128230_1	50/3	1.723	36668.89	3.67
200128230_2	50/3	1.736	37162.35	3.72
200128230_3	50/1	0.839	36245.70	3.62
200128230_4	50/1	0.821	35247.78	3.52

ti result: 36331.18 ppm

ti result: 3.63 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128231	26.0154	28.0252	27.9690	2.80	0.9720

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128231_1	100/3	1.861	153852.15	15.39
200128231_2	100/3	1.903	156269.51	15.63
200128231_3	50/1	1.259	155078.80	15.51
200128231_4	50/1	1.183	151071.48	15.11

co result: 154067.98 ppm

co result: 15.41 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm [dried]	%Ti [dried]
200128231_1	50/3	2.025	41933.45	4.19
200128231_2	50/3	1.978	40704.55	4.07
200128231_3	50/1	0.932	39489.75	3.95
200128231_4	50/1	0.942	40005.73	4.00

ti result: 40533.37 ppm

ti result: 4.05 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128232	27.3291	29.3464	29.2367	5.44	0.9456

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128232_1	100/3	2.129	176003.13	17.60
200128232_2	100/3	2.075	175153.79	17.52
200128232_3	50/1	1.251	163406.75	16.34
200128232_4	50/1	1.316	168175.80	16.82

co result: 170684.87 ppm

co result: 17.07 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128233	25.3688	27.4017	27.3222	3.91	0.9609

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128233_1	100/3	2.198	185069.16	18.51
200128233_2	100/3	2.219	181378.37	18.14
200128233_3	50/1	1.489	192544.70	19.25
200128233_4	50/1	1.347	172298.50	17.23

co result: 182822.68 ppm

co result: 18.28 %

co oxide factor:

co oxide result: 0.00 %

co lot: Manufacturer: Assurance, 28-73COY exp: 2026-10-30

Note(s): sample_3 and sample_4 digested and analyzed by Nickolas Burch.

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128234	25.4696	27.4799	27.3755	5.19	0.9481

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128234_1	100/3	2.261	192762.00	19.28
200128234_2	100/3	2.312	192079.79	19.21
200128234_3	50/1	1.450	179678.93	17.97
200128234_4	50/1	1.517	194564.34	19.46

co result: 189771.26 ppm

co result: 18.98 %

co oxide factor:

co oxide result: 0.00 %

co lot: Manufacturer: Assurance, 28-73COY exp: 2026-10-30

Note(s): sample_3 and sample_4 digested and analyzed by Nickolas Burch.

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128235	31.7284	33.7382	33.6295	5.41	0.9459

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128235_1	100/2	1.668	208535.00	20.85
200128235_2	100/2	1.700	213140.61	21.31
200128235_3	50/1	1.497	195865.67	19.59

co result: 205847.09 ppm

co result: 20.58 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128236	26.4347	28.4458	28.2214	11.16	0.8884

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128236_1	100/2	1.724	225852.56	22.59
200128236_2	100/2	1.646	222898.08	22.29
200128236_4	50/1	1.550	214649.57	21.46

co result: 221133.41 ppm

co result: 22.11 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

Date: 2026-06-04
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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128237	21.1420	23.1600	23.0381	6.04	0.9396

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128237_1	100/2	1.926	239689.23	23.97
200128237_2	100/2	1.879	241989.43	24.20
200128237_3	50/1	1.890	249938.85	24.99

co result: 243872.50 ppm

co result: 24.39 %

co oxide factor:

co oxide result: 0.00 %

co lot: *Manufacturer: Assurance, 28-73COY exp: 2026-10-30*

Note(s): *sample_3 and sample_4 digested and analyzed by Nickolas Burch.*

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$